- ***Automation:*** With cloud management, almost all enterprise operations can run autonomously.
- ***Analysis:*** Enterprises can track cloud workflows, and improve the experience of clients and end users.

How cloud management works

The primary objective of cloud management software is to gather information at regular time intervals. This is necessary for making informed decisions with respect to what to keep in a private or public cloud, and to optimise hybrid and community clouds, respectively.

Public cloud providers typically offer their own software tools for monitoring, securing and managing the cost of their cloud offerings. However, these tools rarely offer insights into performance, and instead stick to basic reporting. Third party tools designed to help manage public cloud services become necessary if organisations are using multiple public clouds that all have their own proprietary cloud management tools.

IT administrators can use private cloud management software tools to allocate resources more efficiently. For instance, an IT manager can use a cloud management tool to install a user-based resource quota to ensure that one user does not overwhelm the server with a large workload request. Administrators can also use data gained from resource monitoring to predict and plan for spikes in resource demands.

Cloud management strategy

Cloud management strategy not just depends on effective tools and automation but also on skilled staff with professional hands-on experience. IT and business teams need to collaborate with each other to understand the requirements and maintain a two-fold strategy for organisational objectives.

IT teams must also test cloud application performance, monitor cloud computing metrics, make critical infrastructure decisions, address patch and security vulnerabilities, and update the business rule sets that drive cloud management. Organisations also must rethink their change management policies for the cloud, where consumption of resources can be much more rapid and spread out vis a vis an on-premise IT environment.

Companies that lack skilled IT staff can seek help from third parties. There are third-party apps that support budget threshold alerts that can notify finance and line of business stakeholders so they can monitor their cloud spending. Cloud brokerages often have a service catalogue and some financial management tools. Cloud spending must be monitored early, when apps go into production. Cloud management training should extend beyond IT and into other departments, such as the supply chain and accounting staff.

Apache CloudStack: An overview

Apache CloudStack is open source cloud computing software for creating, managing and deploying infrastructure cloud services. It makes use of existing hypervisor platforms for performing virtualisation like KVM, VMware vSphere including ESXi, vCenter and XenServer. Apache CloudStack also supports Amazon Web Services and Open Cloud Computing Interface.

The features of Apache CloudStack are listed below.

- ***Effective infrastructure management:*** Apache CloudStack is an effective tool for managing scalable infrastructure where enterprises have tens of thousands of physical servers distributed across geographical locations without any requirement of cluster-level servers. CloudStack enables easy maintenance with respect to updation of software and hardware infra updates without affecting normal operations.
- ***Autonomous configuration management:*** It enables IT admin to automatically configure all sorts of settings in terms of network, security and storage for deployment of virtual machines.
- ***Easy GUI:*** CloudStack offers easy and quickly manageable GUI based on the Web for managing and provisioning the cloud and for managing virtual machines.
- ***API:*** It provides a REST-like API for operating, utilising and managing the cloud.
- ***Less downtime:*** CloudStack has a number of features to increase the availability of the system. The management server itself may be deployed in a multi-node installation where the servers are load balanced. MySQL may be configured to use replication to provide for failover in case of database loss.

  Other features include:
- New modern UI (project primate, technical preview)
- Backup and recovery framework
- Backup and recovery provider for Veeam
- VM ingestion
- CloudStack Kubernetes service
- L2 network PVLAN enhancements
- UEFI support
- KVM rolling maintenance
- Enables direct download for systemVM templates
- Template direct download support for local and SharedMountPoint storages
- VR health checks
- Download logs and diagnostics data from SSVM/CPVM/VRs
- Enables additional configuration metadata to virtual machines

CloudStack: Technical architecture

CloudStack deployment is really flexible and scalable; it can be deployed in simple and primary installations and can even accommodate complex installations.